AI-Enabled Crowdsourced Civic Issue Reporting and Resolution System

ABSTRACT

Civic problems such as potholes, damaged roads, overflowing garbage bins, water leakage, blocked drainage, streetlight failures, and damaged public infrastructure directly affect the daily lives of citizens. Traditional grievance mechanisms in India often rely on physical visits, manual paperwork, or fragmented portals, leading to low visibility and unresolved complaints. Existing digital systems in India, such as the Swachhata-MoHUA mobile application and the CPGRAMS central grievance portal, suffer from critical limitations: CPGRAMS relies on manual routing with long processing cycles, while Swachhata-MoHUA is restricted to a native mobile app without intelligent, automated severity prioritization.

To resolve these challenges, this project proposes the **AI-Enabled Crowdsourced Civic Issue Reporting and Resolution System**, built as a cross-platform, highly accessible **Web Application**. Citizens can seamlessly report civic issues through any modern browser without mandatory app installation, attaching issue descriptions, photographs, and precise GPS location coordinates. The web platform incorporates a multi-modal AI-based severity detection engine inspired by recent research: a hybrid Convolutional Neural Network and Recurrent Neural Network (CNN-RNN) model analyzes uploaded visual evidence, while a Support Vector Machine combined with Natural Language Processing (SVM-NLP) evaluates complaint text. The combined AI severity score automatically prioritizes severe infrastructure hazards on administrative and municipal officer dashboards. Combined with a transparent three-tier architecture (Citizen, Admin/Server, Authority/Department), the system ensures automated verification, real-time lifecycle tracking, and citizen feedback, converting passive grievance submission into an intelligent, transparent e-Governance ecosystem.

Keywords: *Web Application, Civic Issue Reporting, Crowdsourcing, AI Severity Detection,*

CNN-RNN Hybrid Model, SVM-NLP, e-Governance, Indian Civic Tech, CPGRAMS, Swachhata-MoHUA.

1. INTRODUCTION

Urban and semi-urban communities across India face continuous municipal challenges, including potholes, damaged roadways, garbage accumulation, sewage overflows, streetlight outages, and water pipe leakages. Delayed identification and resolution of these issues compromise public safety, urban sanitation, and overall quality of life. While citizens are uniquely positioned to act as real-time sensors of urban infrastructure decay, traditional grievance mechanisms require physical visits, long queues, or tedious paperwork.

Modern e-Governance initiatives have introduced platforms like CPGRAMS (Centralized Public Grievance Redress and Monitoring System) and the Swachhata-MoHUA application. However, these solutions present significant operational gaps:

- **Platform Accessibility Restrictions:** Swachhata-MoHUA requires downloading a native mobile app, creating friction for occasional users or low-storage devices.
- **Lack of Automated Prioritization:** Complaints in CPGRAMS and Swachhata are processed chronologically or routed manually, causing urgent, life-threatening hazards (e.g., open manholes, severe road cave-ins) to sit in queue behind low-priority issues.
- **Manual Routing Delays:** High complaint volumes overwhelm municipal administrators, leading to delayed department assignment and inadequate progress tracking.

To address these shortcomings, this project develops a universal, responsive **Web Application** platform. By implementing a multi-modal AI layer (CNN-RNN for visual severity analysis and SVM-NLP for textual complaint urgency analysis), the system automatically categorizes and prioritizes incoming complaints. The web interface grants universal accessibility across mobile devices, tablets, and desktop computers without requiring native app installation.

2. PROBLEM STATEMENT

Current civic complaint management in India lacks an intelligent, universally accessible, and automated mechanism for collecting, verifying, prioritizing, and resolving civic issues. Manual or semi-digital grievance channels suffer from severe complaint overload, lack of automated prioritization, fragmented communication, and delayed department routing.

Specific problems addressed by this project include:

- **Lack of Smart Prioritization:** Existing Indian portals treat all complaints equally, leading to critical safety hazards being delayed alongside minor maintenance requests.
- **Native App Overhead:** Forcing citizens to install dedicated mobile apps (like Swachhata) reduces civic participation due to storage constraints or device incompatibility.
- **Incomplete Evidence & Manual Processing:** Grievance systems like CPGRAMS lack integrated real-time visual and GPS evidence validation, overburdening administrators with manual verification tasks.

Therefore, a centralized, progressive **Web Application** is required that collects structured reports (images, descriptions, GPS coordinates), applies AI models to estimate complaint severity, and provides transparent workflow tracking for citizens, administrators, and municipal officials.

3. RESEARCH GAP & INDIAN CIVIC TECH ANALYSIS

A critical evaluation of existing Indian civic tech platforms and literature highlights key research gaps:

Platform / Literature Source	Focus Area	Limitations / Flaws	How Our Web Application Resolves It
CPGRAMS Portal (National Grievance System)	Centralized public grievance redressal across ministries.	Manual routing, long response windows (up to 30 days), lack of real-time AI severity detection, no integrated visual evidence severity engine.	Introduces automated AI severity classification (CNN-RNN + SVM-NLP) to route high-urgency issues instantly to nodal department dashboards.
Swachhata-MoHUA App (Ministry of Housing & Urban Affairs)	Sanitation & civic issue reporting.	Native mobile app requirement (App Store/Play Store), restricted platform accessibility, limited automated text/image urgency prioritization.	Built strictly as a responsive **Web Application**, accessible on any browser without installation; features multi-modal AI severity scoring.
Smart Civic Issue Reporting Paper (IJARSCT 2022)	Android app with CNN-RNN & SVM-NLP severity models.	Tied strictly to an Android native app structure and Android NNAPI; lacks web deployment and live Indian e-Governance comparative framework.	Adapts the severity classification algorithms into a decoupled server-side AI layer serving a universal Web Application platform.

4. RESEARCH OBJECTIVES

1. To design and implement a responsive, cross-platform **Web Application** for reporting civic issues across desktop and mobile web browsers without app store dependencies.
2. To enable citizens to submit structured civic reports with images, descriptions, categories, and precise GPS geolocation.
3. To develop an AI severity detection layer using a hybrid CNN-RNN model for image analysis and an SVM-NLP model for text analysis to estimate complaint severity automatically.
4. To implement a priority-based queue management engine that ranks complaints based on AI severity scores, issue categories, and administrative rules.
5. To construct an administrative module for verification, deduplication, and department routing, along with an authority/department dashboard for updating status and closing tickets.
6. To provide real-time status tracking and post-resolution feedback for citizens.
7. To evaluate the web platform in terms of usability, page response times, workflow efficiency, and ML classification performance.

5. LITERATURE REVIEW

Research in e-Governance demonstrates that crowdsourced citizen reporting significantly improves municipal asset maintenance. Early platforms such as *MarkaSpot* and *AppCivist* demonstrated the utility of geo-tagged photography and open web standards for public engagement.

In the Indian context, platforms like CPGRAMS and the Swachhata-MoHUA app established digital grievance channels. However, as highlighted by recent civic tech studies (e.g., Jan Suvidha 2025), large-scale government portals suffer from fragmented interfaces, delayed response times, and manual complaint routing.

The baseline study by Walwadkar et al. (IJARSCT 2022, DOI: 10.48175/IJARSCT-2659)

proposed integrating machine learning into civic reporting. They introduced a hybrid CNN-RNN architecture for visual severity classification and an SVM classifier with NLP for textual description urgency. Their approach demonstrated that combining feature extraction with sequence modeling enhances image classification accuracy compared to standalone CNNs. This project adopts and refines these algorithmic concepts, embedding them into a web-based backend architecture to overcome the mobile-only constraints of prior work.

6. PROPOSED SYSTEM ARCHITECTURE

The system is designed as a **Three-Tier Web Architecture** adhering to the Model-View-Controller (MVC) design pattern:

6.1 Client / Presentation Layer (Web Portal)

- Built using modern web technologies (HTML5, CSS3/Tailwind CSS, JavaScript/React web interface).
- Fully responsive UI optimized for mobile web browsers, laptops, and desktop displays.
- Captures GPS geolocation via Web Location APIs, handles photo uploads, and presents status tracking dashboards.

6.2 Application & AI Processing Server Layer

- Web API endpoints handle user authentication, report submission, and administrative routing.
- **AI/ML Severity Engine:** Receives uploaded media and text descriptions. Executes CNN-RNN image classification and SVM-NLP text evaluation asynchronously to return a unified severity index.

6.3 Database & Storage Layer

- Centralized relational/document database storing user profiles, complaint records, geo-coordinates, status logs, AI severity scores, and audit trails.

7. AI-BASED SEVERITY DETECTION METHODOLOGY

To prioritize high-risk complaints over routine maintenance requests, the system implements a dual AI layer:

7.1 Image-Based Severity Detection (CNN-RNN Hybrid)

- **CNN Component:** Extracts spatial feature maps from uploaded civic complaint images (e.g., depth of potholes, size of garbage heaps, broken structural concrete).
- **RNN Component:** Processes spatial sequence representations to model severity hierarchies.
- **Output:** Categorizes visual severity into levels (e.g., Low, Moderate, High, Severe).

7.2 Text-Based Severity Detection (SVM with NLP)

- **NLP Preprocessing:** Tokenization, stop-word removal, and TF-IDF term vectorization on complaint descriptions.
- **SVM Classifier:** Evaluates urgency keywords (e.g., "danger", "accident", "overflowing", "collapse") to assign a textual severity rating.

7.3 Multi-Modal Severity Score Integration

The system computes a combined Priority Rank (P) based on image severity score, text severity score, tuned weighting factors, and baseline category hazard weights.

8. FUNCTIONAL MODULES & COMPLAINT WORKFLOW

8.1 Functional Modules

- **Citizen Web Portal:** Registration, complaint submission, image upload, browser GPS location sharing, real-time ticket tracking, feedback submission.
- **AI Severity Module:** Asynchronous processing of visual and textual complaint data to generate priority scores.

- **Admin Management Portal:** Review incoming complaints, deduplicate reports, verify AI severity flags, and assign complaints to municipal departments.
- **Authority / Department Portal:** View assigned tasks sorted by priority rank, update progress status, upload post-resolution evidence photos, and close tickets.
- **Notification Engine:** Generates web push alerts and SMS/email updates to citizens upon status transitions.

8.2 Complaint Lifecycle / Status Model

The complaint moves through six transparent stages:

1. **Reported:** Citizen submits issue via web portal.
2. **Approved / Verified:** Administrator reviews and confirms validity.
3. **Assigned:** System/Admin routes complaint to designated municipal department.
4. **In Progress:** Authority team initiates field action.
5. **Resolved:** Field team uploads resolution proof and updates status.
6. **Closed:** Case completed following citizen verification/feedback or administrative audit.

9. EVALUATION METRICS

The proposed system will be evaluated across ML performance and web platform operational metrics:

- **ML Classification Metrics:** Accuracy, Precision, Recall, F1-Score, and Confusion Matrix across severity categories.
- **Web System Performance:** Page load times, API response latency, concurrent session handling, and mobile browser compatibility.
- **Workflow Efficiency:** Average time elapsed from complaint submission to department assignment and final resolution comparison against baseline manual channels.

10. EXPECTED OUTCOMES

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  A cross-platform **Web Application** providing friction-free civic reporting across all modern web browsers.
- Automated prioritization of life-threatening infrastructure hazards via multi-modal AI analysis.
- Reduction in administrative routing overhead and complaint verification bottlenecks.
- Transparent tracking for citizens, increasing accountability in public administration.
- Structured database analytics enabling municipal authorities to identify recurring civic failure zones.

11. LIMITATIONS

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  Data Quality Dependence: AI severity performance depends on image resolution, lighting conditions, and textual description clarity.
- **GPS Precision:** Web geolocation accuracy varies based on browser permissions, network signal strength, and hardware capabilities.
- **Physical Resolution Dependency:** Digital prioritization streamlines administrative routing, but physical repair relies entirely on municipal department field execution.

12. FUTURE SCOPE

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  Automated Image Deduplication: Using perceptual hashing and visual similarity models to group multiple reports of the same physical pothole or garbage dump.
- **Multilingual NLP Support:** Expanding textual severity analysis to support regional Indian languages (Hindi, Marathi, Tamil, etc.).
- **Heatmap Analytics Dashboard:** Generating interactive municipal maps displaying high-density civic issue zones for predictive infrastructure maintenance.
- **Municipal ERP Integration:** Direct API integration with official government portals for

seamless ticket escalation.

13. CONCLUSION

The **AI-Enabled Crowdsourced Civic Issue Reporting and Resolution System** bridges the gap between citizens and municipal authorities by combining universal web accessibility with intelligent AI prioritization. By resolving the key limitations of existing Indian platforms—such as the native app barrier of Swachhata-MoHUA and the manual routing delays of CPGRAMS—this web application provides a scalable, transparent, and efficient foundation for modern e-Governance and smart city infrastructure management.

14. REFERENCES

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